

Reanalysis of "Community estimate of global glacier mass changes from 2000 to 2023"

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Introduction

The study by Zemp *et al.* [1] integrated observations across multiple methods into a combined estimate of global glacier mass change. The estimates are the result of the Glacier Mass Balance Intercomparison Exercise (GlaMBIE), a community project that collected glacier mass changes from 35 research teams using four different observation methods: glaciological field measurements, DEM differencing, altimetry, and gravimetry. Zemp *et al.* [1] found that global glaciers lost approximately $-6,542 \pm 387$ Gt (gigatons) of mass between 2000 and 2023 (-273 ± 16 Gt/yr), representing roughly 5% of total glacier mass globally, with the rate of loss accelerating by $36 \pm 10\%$ from the first to the second half of the observation period.

To derive these estimates, Zemp *et al.* [1] homogenized the datasets in space, time, and unit. Annual variability is separated from long-term trends before combining methods into regional and global estimates. Their approach allowed for high-temporal glaciological observations to complement the long-term accuracy of DEM differencing, which enables annual estimates across a long time period and a comparison of observation methods.

Our reanalysis confirmed the robustness of Zemp *et al.* [1] at a global scale whereas regional results are more sensitive to methodology and data selection. In our reanalysis we investigated several points of interest: the sensitivity of the estimates to the choice of processing methods, the glacier area inventory version used, the assumption of linear glacier area changes, and the exclusion of datasets.

Methods

We tested 4 different approaches for trend extraction and temporal homogenization that were already implemented in the GlaMBIE model. Global estimates changed up to 0.9% (equivalent to 58 Gt) depending on the chosen method, while the difference in regional estimates ranged from -10 to +12% (Table 1). Zemp *et al.* [1] chose a regression approach for inter-annual trend extraction, which is less sensitive to anomalies than the alternative approach (start-vs-end). For temporal alignment, they assumed a linear change, but it remains unclear whether the more complex analytical model of seasonal homogenization would yield more accurate results. Although the regional differences showed uncertainty in the methodological approach, the global results stayed consistent and we found the authors' chosen methods appropriate.

Relative change

Expressing glacier changes only in absolute terms, as the authors did, can result in an underestimation of the acceleration of mass loss. We found that six regions show significant decreasing trends when the change is evaluated relative to the remaining glacier mass, whereas only three do for absolute mass changes. Although the global trend remains nearly unchanged during the observed period, due to the relatively small overall change of 5%, relative change should still be considered, especially on a regional level, as otherwise we risk the acceleration of glacier melt going unnoticed.

Glacier area

The authors computed the regional glacier area timeseries using a snapshot regional glacier area from the Randolph Glacier Inventory (RGI 6.0) and the static regional glacier area-change rates (% per year) compiled from IPCC AR5, extended with additional literature from Zemp *et al.* (RGI Consortium [2]; IPCC [3]; Zemp *et al.* [4]). We had three comments about this.

Firstly, RGI 6 got updated to RGI 7 after GlaMBIE launched (RGI Consortium [5]). We tested the impact of the update and found that, while the global impact is small, using the updated database can lead to a change in glacier mass loss of up to 22 Gt, with relative differences of up to 23.7% on a regional level (table 1). As such, while we do not fault the authors for not yet having done so, it is imperative that the updated inventory is implemented in the GlaMBIE model.

Secondly, by using a static change rate, the resulting glacier area time series is linear. We investigated the effects of accelerating glacier area loss on the model by running two additional models with an increasing change rate (Models 2012 and 2023). Globally, the effect was small. However, on a regional level, we found absolute and relative differences of up to 1.13 m w.e. (meters water equivalent) and 4.42%. Although the effect of including non-linearity in the area shrinkage is small, including it does not increase computation time and, as such, updating GlaMBIE with a realistic glacier area shrinkage should be considered.

Lastly, Zemp *et al.* [1] are unclear about how exactly the glacier area change rates were calculated. Although they are similar to the area-weighted average of the IPCC AR5 rates, they are not the same and, since the numbers by Zemp *et al.* [4] are not accessible to the public, the cause of this difference remains uncertain. Furthermore, while these rates are used to account for glacier area shrinkage, when converting their results from m w.e. to Gt, they first remove this effect, and thus use only the reference glacier area in their conversion, resulting in glacier shrinkage only impacting their results in m w.e. while having little to no effect on their results in Gt. It is unclear why they do this, as it is not explained in their paper or the supplementary information, but it is clearly a deliberate decision. As such, more clarity on the glacier area calculations is needed; both on how the glacier area change rates were calculated and on the reason for excluding shrinkage from the Gt conversion.

	Unit	Abs Δ		Rel Δ (%)		Spearman r	
		Global	Regional Range	Global	Regional Range	Global	Regional Range
Methods							
Regression + proportional	Gt	0.00	0.00 to 0.00	0.00	0.00 to 0.00	1.00	1.00 to 1.00
Regression + seasonal homog.	Gt	26.12	-12.66 to 26.34	0.40	-4.08 to 6.16	1.00	0.92 to 1.00
Start-End + proportional	Gt	31.98	-19.03 to 73.79	0.49	-9.56 to 11.70	1.00	1.00 to 1.00
Start-End + seasonal homog.	Gt	58.11	-20.05 to 70.71	0.89	-10.07 to 11.21	0.99	0.92 to 1.00
Glacier Area							
RGI 7	Gt	-0.42	-15.75 to 22.43	-0.01	-9.67 to 23.75	1.00	1.00 to 1.00
Model 2012	m w.e	-0.10	-1.13 to -0.01	-1.03	-4.42 to -0.12	1.00	0.99 to 1.00
Model 2023	m w.e	0.11	0.02 to 0.84	1.15	0.27 to 4.67	1.00	1.00 to 1.00
Datasets							
All	Gt	50.31	-44.96 to 44.80	0.77	-99.48 to 139.64	0.97	0.01 to 1.00
Including gravimetry	Gt	79.56	-38.58 to 44.80	1.22	-99.48 to 139.64	0.95	0.01 to 1.00
All except gravimetry	Gt	-41.45	-18.43 to 3.55	-0.63	-13.69 to 0.86	0.99	0.94 to 1.00

Table 1: Summary of cumulative glacier mass change variations and correlations across glacier area, method and dataset adjustments compared to the results from Zemp *et al.* [1].

Datasets

The biggest variability in our reanalysis arises from dataset inclusion and exclusion. Out of 233 datasets, 38 were excluded. Global estimated cumulative changes differed up to +1.2%, while regionally, the changes ranged from -99% to +139%, depending on dataset selection (Table 1). The largest impact was on small-area glacier regions. The exclusion of non-gravimetry datasets led to much smaller variations (-14% to +1%). We agree with the exclusion of gravimetry datasets due to problematic trends and large uncertainties, though we note that the exclusion of most gravimetry datasets weakens the main novelty claimed by Zemp *et al.* [1] - incorporating multiple observational methods. Some regions are solely represented by DEM-differencing. Additionally, we faced technical problems when trying to include all datasets. Overall, increased transparency about the reason for the exclusions would strengthen the papers credibility due to the magnitude by which dataset selection affects the results.

Conclusion

To conclude, although Zemp *et al.* [1] provide a convincing global synthesis of glacier mass loss, the regional estimates are a lot more uncertain and should be interpreted with greater caution. This also highlights the limitations of the current state of glacier monitoring, which future research should address with improved observations. Furthermore, more clarity on the glacier area calculations and the reasoning behind excluding shrinkage from the Gt conversion is needed, as well as the reasons for the dataset exclusion.

References

1. Zemp, M. *et al.* Community estimate of global glacier mass changes from 2000 to 2023. *Nature* **639**, 382–388 (2025).
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3. IPCC. *Climate Change 2013: The Physical Science Basis*. (eds Stocker, T. F. *et al.*) (Cambridge University Press, 2013).
4. Zemp, M. *et al.* Global glacier mass changes and their contributions to sea-level rise from 1961 to 2016. *Nature* **568**, 382–386 (2019).
5. RGI Consortium. *Randolph Glacier Inventory - A Dataset of Global Glacier Outlines, Version 7* 2023.